1) STUDY OF VARIOUS NETWORK COMMANDS USED IN LINUX AND WINDOWS.

1) arp -a

=> Address resolution Protocol.

=> It show IP address and MAC address of Router.

2) host name:

=> Displays name of the computer.

3) IPconfig/all:

=> displays detailed Configuration information about your TCP/IP connection including router, DNS, and type of ethernet adapter.

4) nbstat -a:

=> command helps to solve problem with NetBIOS name Resolution.

5) net stat:

=> Display variety of statistics about a Computers active TCP/IP connections.

6) DSlookup:

=> DNS Lookup in Linux.

=> display DNS details such as the IP address of a particular computer, the MX records for a domain.

7) Ping:

TO test connectivity blw a node

1) Ping hostname.

2) # Ping id address.

3) # Ping fully available domain name.

2) If config:

JHP if config command was a staple in many sysadmin's tool belt for configuring and trouble shooting networks.

a) To show statistics including each hop (hostname) with time and loss %.

b) To show numeric IP address.

ex: google.com

c) show numeric IP address and hostnames

#mtr -b google.com.

IP:

The boss ip command can show address information, manipulate routing, plus display network various devices and tunnes.

a) To show the IP address assigned to an interface on your server.

ip address show.

b) To assign an IP to an interface.

for ex: # ip address add 192.168.1.254/24 dev enP003.

c) To delete an IP on an interface

#ip address dd 192.168.1.254/2

dev

d) Alter the status of the interface by the bringing the interface eth0 online.

e) Alter the status of the interface by the bringing the interface.

f) Add a default route via The local gateway 192.168.1 That can attached on a device eth0.

g) Add a route to 192.168.1.0/2 That can be reached on device eth0.

#ip route add 192.168.1.0/24 dev eth0.

h) delete the route.

to dump'.

=> Command for capturing and displaying packets.

a) to install tcpdump

#dnf - y tcpdump

b) To know which interface tcp dump can use.

#tcp dump -D.

c) To capture traffic on ethO, we can initiate that with tcpdump -i

d) # tcp dump -i eth0 -c10 => capture. traffic to and from one host.

e) #tcpdump -i eth0 sc host 8.8.8.8 => coming from 8.8.8.8

f) #tcpdump -i eth0 port 53 To capture specific port.

g) #tcp dump -i eth0 -c 10 host www.google.com. and port 443 to capture HTTPS path.c.